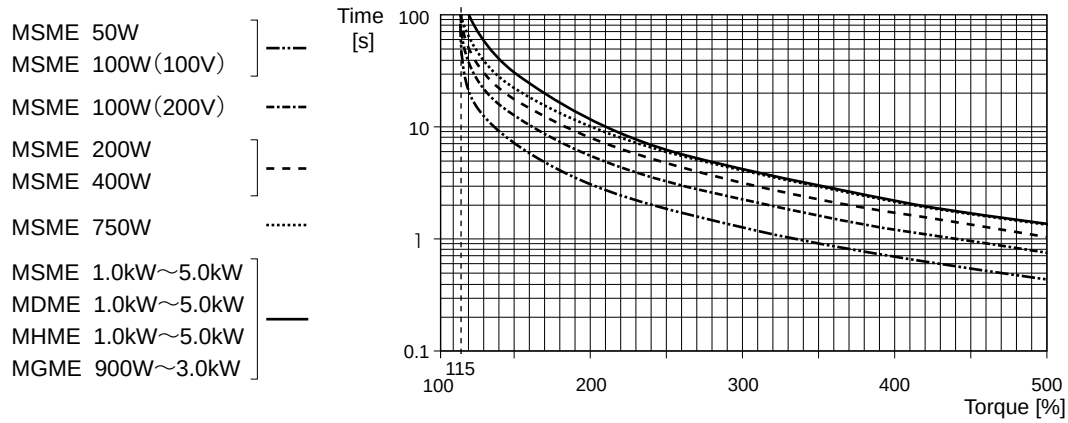
Time characteristics of Error code No.16.0 (Overload protection)



Caution Use the motor so that actual torque stays in the continuous running range shown in “S-T characteristic” of the motor. For the S-T characteristics, see P.7-56 Motor characteristics (S-T characteristics).

Software Limit Function (Error code No.34.0)

1) Outline

You can make an alarm stop of the motor with software limit protection (Error code No.34) when the motor travels exceeding the movable range which is set up with Pr5.14 (Set up of software limit) against the position command input range.

You can prevent the work from colliding to the machine end caused by motor oscillation.

2) Applicable range

This function works under the following conditions.

	Conditions under which the software limit works
Control mode	• Position control, Full-closed control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

3) Cautions

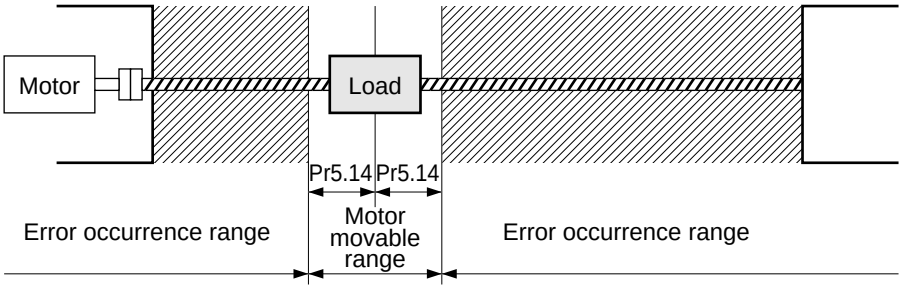
- This function is not a protection against the abnormal position command.
- When this software limit protection is activated, the motor decelerates and stops according to Pr5.14 (Sequence at alarm).
  The work (load) may collide to the machine end and be damaged depending on the load during this deceleration, hence set up the range of Pr5.14 including the deceleration movement.
- This software limit protection will be invalidated during the trial run and frequency characteristics functioning of the PANATERM.

Related page • P.4-44, 4-45 “Details of parameter”

4) Example of movement

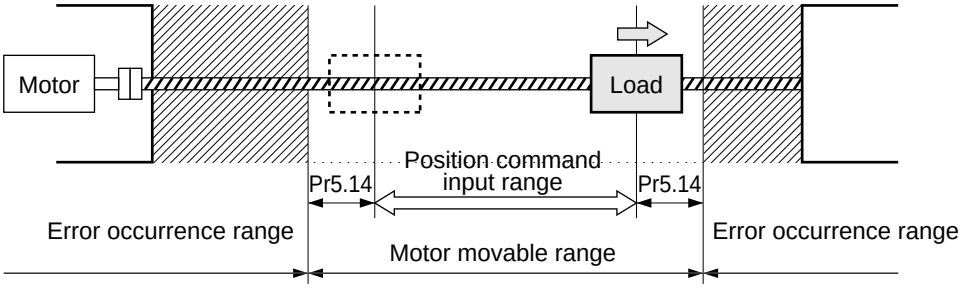
(1) When no position command is entered (Servo-ON status),

The motor movable range will be the travel range which is set at both sides of the motor with Pr26 since no position command is entered. When the load enters to the Err34 occurrence range (oblique line range), software limit protection will be activated.



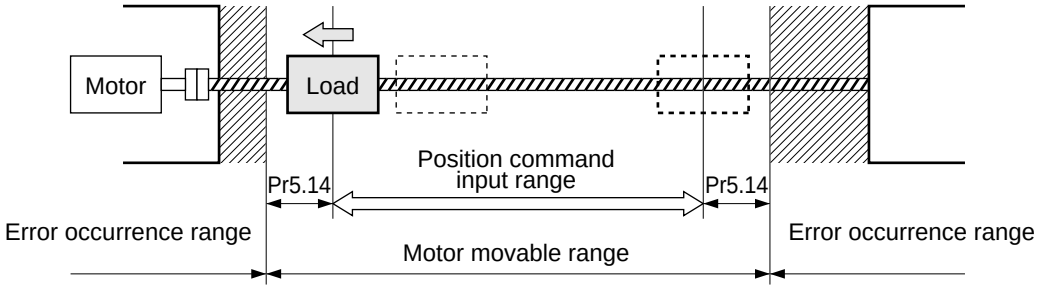
(2) When the load moves to the right (at Servo-ON),

When the position command to the right direction is entered, the motor movable range will be expanded by entered position command, and the movable range will be the position command input range + Pr26 setups in both sides.



(3) When the load moves to the left (at Servo-ON),

When the position command to the left direction, the motor movable range will be expanded further.



5) Condition under which the position command input range is cleared

The position command input range will be 0-cleared under the following conditions.

- when the power is turned on.
- while the position deviation is being cleared (Deviation counter clear is valid, Pr66 (Sequence at over-travel inhibition) is 2 and over-travel inhibition input is valid.)
- At the beginning and ending of trial run via communication.

Related page • P.4-42... “Details of parameter”

2. Setup of gain pre-adjustment protection

Before starting gain adjustment, set the following parameters based on the conditions of use, to assure safe operation.

1) Setup of over-travel inhibit input

By inputting the limit sensor signal to the driver, the bumping against mechanical end can be prevented. Refer to interface specification, positive/negative direction over-travel inhibit input (POT/NOT). Set the following parameters which are related to over-travel inhibit input.

Pr5.04 Setup of over-travel inhibit input

Pr5.05 Sequence at over-travel inhibit

Related page P.3-38 (POT/NOT), P.4-42 (Pr5.04, Pr5.05)

2) Setup of torque limit

By limiting motor maximum torque, damage caused by failure or disturbance such as bite of the machine and collision will be minimized. To apply standardized limit through parameters, set Pr0.13 The 1st torque limit.

If the torque limit setup is lower than the value required during the actual application, the following two protective features will be triggered: over-speed protection when overshoot occurs, and excess positional deviation protection when response to the command delays.

By allocating the torque in-limit output (TLC) of interface specification to the output signal, torque limit condition can be detected externally.

Related page P.3-45 (TLC), P.4-10 (Pr0.13), P.4-47 (Pr5.21)

3) Setup of over-speed protection

Generates Err26.0 Over-speed protection when the motor speed is excessively high.

Default setting is the applicable motor maximum speed [r/min] × 1.2.

If your application operates below the motor maximum speed, set Pr5.13 Setup of over-speed level by using the formula below.

Pr5.13 Setup of over-speed level = Vmax × (1.2 to 1.5)
Vmax: motor maximum speed [r/min] in operating condition
Factor in ( ) is margin to prevent frequent activation of over-speed protection.

When running the motor at a low speed during initial adjustment stage, setup the over-speed protection by multiplying the adjusting speed by a certain margin to protect the motor against possible oscillation.

Related page P.4-45 (Pr5.13)

(Continued ...)

4) Setup of the excess positional deviation protection

During the position control or full-closed control, this function detects potential excessive difference between the positional command and motor position and issues Err24.0 Excess positional deviation protection.

Excess positional deviation level can be set to Pr0.14 Setup of positional deviation excess. The deviation can be detected through command positional deviation [pulse (command unit)] and encoder positional deviation [pulse (encoder unit)], and one of which can be selected by Pr5.20 Position setup unit select. (See the control block diagram.)

Default setting is 100000[pulse (command unit)].

Because the positional deviation during normal operation depends on the operating speed and gain setting, fill the equation below based on your operating condition and input the resulting value to Pr0.14.

• When Pr5.20 = 0 (detection through command positional deviation)

Pr0.14 Setup of positional deviation excess = Vc/kp × (1.2 to 2.0)
Vc: maximum frequency of positional command pulse [pulse (command unit)/s]
Kp: position loop gain [1/s]
Factor in ( ) is margin to prevent frequent activation of excess positional deviation protection.

Note 1) When switching position loop gain Kp, select the smallest value for calculation.
Note 2) When using the positional command filter and damping control, add the following values.
Positional command smoothing filter: Vc × filter time constant [s]
Positional command FIR filter: Vc × filter time constant [s]/2
Damping control: Vc/(π × damping frequency [Hz])

• When Pr5.20 = 1 (detection through encoder positional deviation, full-closed positional deviation)

Pr0.14 Setup of positional deviation excess = Ve/Kp × (1.2 to 2.0)
Ve: maximum operation frequency [pulse/s] in encoder unit or full-closed unit
Kp: position loop gain [1/s]

Note 3) When switching position loop gain Kp, select the smallest value for calculation.
Note 4) When Pr5.20 = 1, setups of positional command filter and damping control have no effect.

Related page P.4-10 (Pr0.14), P4-47 (Pr5.20)

5) Setup of motor working range

During the position control or full-closed control, this function detects the motor position which exceeds the revolutions set to Pr5.14 Motor working range setup, and issues Err34.0 Motor working range protection.

Related page P.4-15 (Pr5.14)

6) Setup of hybrid deviation excess error protection

At the initial operation with full-closed control, operation failure may occur due to reverse connection of external scale or wrong external scale division ratio.

To indicate this type of defect, Err25.0 Hybrid deviation excess error protection is issued when the deviation of motor position (encoder unit) and load position (external scale unit) exceed Pr3.28 Setup of hybrid deviation excess.

Default setting is 16000 pulse (command unit). Because the deviation in normal operation varies with the operation speed and gain setup. Add a margin to this setting according to your operating condition.

Related page P.4-31 (Pr3.28)